Course Review and Mini-Project Demonstration

Communication Technologies for the Internet of Things, Lecture 14

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Agenda

- Course review by blocks
- How to study for the written exam
- What is assessed in the mini-project demonstration
- Structure of the demonstration and common mistakes

The Course Path (1/2)

- **Block 1, Lectures 1 and 2:** IoT reference architecture, sensors, analogue-to-digital conversion and interrupts.
- **Block 2, Lectures 3 to 7:** short- and long-range wireless technologies: LPWAN, Bluetooth, BLE and Zigbee.
- **Block 3, Lectures 8 to 10:** IP networks for IoT: IPv4, NAT, IPv6, 6LoWPAN, 6TiSCH and routing.

The Course Path (2/2)

- **Block 4, Lectures 11 to 13:** application protocols and middleware: HTTP, CoAP, XMPP, MQTT, AMQP and integration with Zigbee2MQTT.
- The order follows the stack, from the bottom up: the signal, the network, the application and the middleware.
- The mini-project runs through all four blocks in a single system: node, communication, gateway and broker.

Map of the Course

Block 4 **Application and middleware**	Lectures 11-13: CoAP, MQTT, AMQP, Zigbee2MQTT
Block 3 **IP networks for IoT**	Lectures 8-10: IPv6, 6LoWPAN, 6TiSCH, RPL
Block 2 **Wireless technologies**	Lectures 3-7: LPWAN, Bluetooth, BLE, Zigbee
Block 1 **Architecture and sensors**	Lectures 1-2: architecture, sensors, ADC, interrupts

Block 1 Review: Architecture and Sensors

- The reference architecture separates devices, communication, platform and application.
- An analogue sensor requires conversion: the converter's resolution determines the smallest step measured.
- Interrupts let the system react to events without continuously polling the sensor, and save energy.
- Be able to compute a converter's resolution and justify the choice between polling and interrupts.

Block 2 Review: Wireless Technologies

- Each technology occupies a trade-off between range, data rate and consumption.
- LPWAN favours range and consumption, with very low data rates.
- BLE favours consumption at short range, with the GATT model of services and characteristics.
- Zigbee forms low-power mesh networks, with a coordinator, routers and end devices.
- Be able to choose a technology for a scenario and justify the choice against the alternatives.

Block 3 Review: IP Networks for IoT (1/2)

- IPv4 ran out. NAT postponed the problem at the expense of the end-to-end model; IPv6 solves it.
- IPv6 brings link-local addresses, stateless autoconfiguration and Neighbor Discovery instead of ARP.
- 6LoWPAN compresses headers and fragments packets to fit in 127-octet frames.

Block 3 Review: IP Networks for IoT (2/2)

- TSCH uses slotframes and channel hopping to give timing guarantees and save energy.
- RPL organises the network in a DODAG and chooses parents by objective function, with the Trickle Timer controlling signalling.
- Be able to compute a frame budget and explain the sequence of an autoconfiguration.

Block 4 Review: Application and Middleware (1/2)

- HTTP is heavy for constrained devices. CoAP keeps the REST model over UDP, with confirmable or non-confirmable messages.
- XMPP offers federation and presence, and belongs on the gateway side.
- Middleware decouples producers and consumers in space, in time and in synchronisation.

Block 4 Review: Application and Middleware (2/2)

- MQTT offers publish-subscribe, three QoS levels, retained messages and a last will.
- AMQP offers queues, exchanges, transactions and persistence, and is aimed at enterprise integration.
- Zigbee2MQTT shows the bridge between the Zigbee world and the middleware.
- Be able to justify the choice between MQTT, AMQP and CoAP and explain the effect of QoS.

How to Study for the Written Exam

- Use the course notes, which follow the slides one by one, and answer the check questions without looking.
- Practise the calculations: frame budget, converter resolution, EUI-64 address, solicited-node address, rank by ETX.
- Practise the comparisons: wireless technologies, HTTP against CoAP against XMPP, MQTT against AMQP.
- Practise the justifications: given a scenario, choose and justify with range, energy and reliability requirements.

What Is Assessed in the Demonstration (1/2)

- The demonstration is oral, in class, and there is no report: the working system is the document.
- Each group member must be able to explain any part of what is presented.
- What is assessed is the wireless communication chosen and justified, the MQTT broker and/or CoAP server, and the consumers.

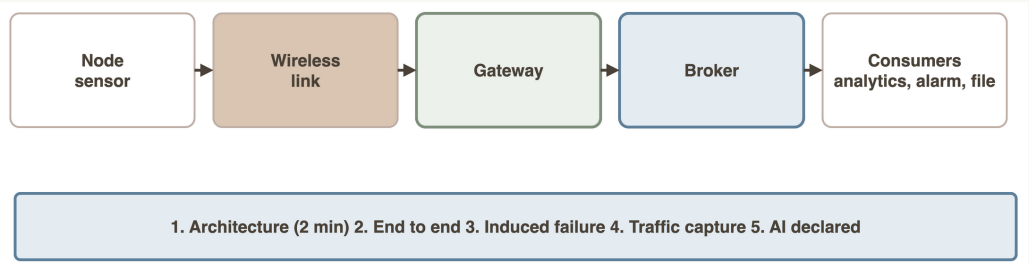
What Is Assessed in the Demonstration (2/2)

- In MQTT: topic hierarchy, wildcards, justified QoS, retained messages, last will, keep-alive and authentication.
- In CoAP: resources reachable by GET, confirmable and non-confirmable messages, and resource observation.
- In operation: analytics, alarms, serialisation, behaviour under failure and a traffic capture that supports a claim.

Structure of the Demonstration

- Start with the architecture in two minutes: node, gateway, broker and consumers.
- Show the system working end to end, before explaining the details.
- Cause a failure, for example switch off the node or the gateway, and show the reaction.
- Show a traffic capture that supports a claim, such as the size of a message or the QoS.
- Declare the use of AI and explain what was accepted and what was corrected.

The Demonstration Path



The demonstration path follows the mini-project chain

Common Mistakes

- Topics with no hierarchy, or with secrets and personal identifiers in the name.
- Credentials written in the code, instead of environment variables or configuration files.
- QoS chosen without justification, or always at the maximum.
- A demonstration that only works once, with no plan B for network or hardware failure.
- Claims with no measurement: when saying it saves energy or bandwidth, show the number.

Summary

- The course teaches how to choose and justify: technology, network, protocol and middleware.
- The mini-project integrates the four blocks in a single system, and the demonstration shows that it works and that it is understood.
- The written exam checks understanding of the principles, with calculations, comparisons and justifications.

Thank you.

Questions and comments